

Report
Of
INTERN APP: A ROLE-BASED USER MANAGEMENT SYSTEM

Submitted
To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: **2410030800**
Name of Student: **APOORVA GUPTA**
Batch: **2026-27**

CANDIDATE'S DECLARATION

I, **Apoorva Gupta**, do hereby declare that the internship report titled "**Intern App: A role-based user management system**" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: **Apoorva Gupta**

Roll No.: **2410030800**

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout this project. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

Student Name: **Apoorva Gupta**

Roll No.: **2410030800**

INTERNSHIP COMPLETION CERTIFICATE



TABLE OF CONTENTS

Chapter No.	Chapter Name	Page Nos.
	Cover Page	
1.	Candidate's Declaration	
2.	Acknowledgement	
3.	Internship Completion Certificate	
4.	Project Description 4.1 Introduction 4.2 Organization / Institution Profile 4.3 Problem Statement 4.4 Project Objectives 4.5 Scope of the Project 4.6 Technologies and Tools Used 4.7 System Architecture 4.8 Methodology 4.9 Expected Outcomes 4.10 Certificates of completion and other communication mail proof	
5.	Bibliography / References	

4. PROJECT DESCRIPTION

4.1 Introduction

Web-based information systems form the backbone of most modern digital services, from small institutional portals to large-scale enterprise platforms. As part of the practical, hands-on component of the B.Tech CSE curriculum with specialization in Data Science, this project was undertaken to apply classroom knowledge of HTML, CSS, SQL and PHP to build a complete, working, database-driven web application from the ground up.

The project, named “Intern App”, is a role-based user management system that allows individuals to register, authenticate, and interact with the platform according to their assigned role — either as a regular user or as an administrator with elevated privileges. The intent behind choosing this project was to gain first-hand, practical exposure to the core concerns of real-world web development: session management, authentication security, database design, server-side validation, and role-based access control (RBAC), all of which are foundational concepts expected of any software or data engineering professional.

4.2 Organization Profile

This internship was undertaken at Physics Wallah (PW), one of India's leading educational technology companies, headquartered in Sector 62, Noida, Uttar Pradesh. PW was founded in 2016 by Alakh Pandey as a YouTube channel offering free physics lectures for students preparing for the Joint Entrance Examination (JEE), and was formally established as an ed-tech company in 2020 along with co-founder Prateek Maheshwari.

Since then, PW has grown into a large-scale, hybrid education platform serving students across the country through online live classes, recorded video courses, test series, and a network of offline coaching centres known as "Vidyapeeth." Its offerings span school education (classes 6–12), competitive exam preparation (JEE, NEET, and other entrance tests), a skilling and higher-education vertical (PW Skills, PW Institute of Innovation), and additional ventures acquired or launched under the PW group. The company reached unicorn status in 2022 after a funding round that valued it above one billion dollars, and has since scaled to a workforce of over 30,000 employees, later

going public and listing on Indian stock exchanges. PW's core mission is to make quality education affordable and accessible to students from all backgrounds, including those in smaller towns and underserved regions.

4.3 Problem Statement

Many small organizations, student clubs, and academic departments require a lightweight system to manage member or user records — registering people, authenticating logins, and distinguishing between users who can only view their own data and administrators who can manage everyone's data. Building such a system from scratch, while keeping it secure (hashed passwords, prepared SQL statements, server-side authorization checks) and usable, was identified as a well-scoped, practical problem suited to demonstrate core web-development and database skills within a limited time frame.

4.4 Project Objectives

- To design and implement a normalized MySQL database schema for storing user records with distinct roles.
- To implement secure user registration and login using PHP with hashed passwords (bcrypt via `password_hash()`).
- To enforce role-based access control on the server side, not merely by hiding UI elements.
- To build a complete CRUD (Create, Read, Update, Delete) interface for administrators to manage user accounts.
- To allow users to view and edit their own profile, including uploading a profile picture.
- To apply the concepts of sessions, prepared statements, and input validation learned in coursework to a working end-to-end system.

4.5 Scope of the Project

The scope of this project is intentionally limited to a normal, small-scale deployment suited for a single institution, department, or small team — rather than a large-scale, high-traffic commercial system. It covers user authentication, two roles (admin and

user), profile management with image upload, and administrative CRUD operations on user records. It does not cover features such as multi-factor authentication, email verification workflows, password-reset-via-email, load balancing, caching layers, or REST API exposure, all of which would be relevant for a production-grade, enterprise-scale deployment but fall outside the intended scope of this academic exercise.

4.6 Technologies and Tools Used

The following technologies and tools were used in the design, development and testing of the project:

Category	Technology / Tool
Front-end	HTML5, CSS3
Back-end	PHP 8 (procedural, mysqli extension)
Database	MySQL 8, managed via phpMyAdmin
Local Server	WAMP Server (Apache, MySQL, PHP on Windows)
Security	password_hash() / password_verify() (bcrypt), prepared statements (mysqli bind_param)
Version/Editor	Visual Studio Code

Table 4.1: Technologies and Tools Used

4.7 System Architecture

The system follows a classic three-tier web architecture:

- Presentation Layer: HTML5 and CSS3 render the user interface — registration, login, dashboards, and user-management forms — shared across pages via reusable header, footer and navbar includes.
- Application Layer: PHP handles all business logic — session management, input validation, authentication, authorization (role checks), and CRUD operations — organized into logical modules (config.php, functions.php, auth.php) that every page includes through a single bootstrap file.
- Data Layer: A MySQL database stores all persistent data in a single normalized users table (id, name, email, password, role, profile picture, created_at), accessed exclusively through parameterized queries to prevent SQL injection.

Every protected page performs its own server-side authorization check (require_login() / require_admin()) before rendering, ensuring that access control cannot be bypassed

by navigating directly to a URL — a deliberate design decision to avoid relying on client-side/UI-only restrictions.

4.8 Methodology

The project was developed following an incremental, module-first methodology:

- Requirement identification and database schema design (defining the users table and its constraints).
- Core infrastructure setup: database connection (config.php), shared helper functions (functions.php), and session/role guard functions (auth.php).
- Authentication module: registration with server-side validation and duplicate-email checks, login with hashed password verification, and logout with full session destruction.
- Role-based dashboards: a distinct view and permitted actions for admin users versus regular users.
- Admin CRUD module: add, list, edit and delete user records, with confirmation prompts before destructive actions.
- Iterative testing: manual functional testing of each module (registration, login, role redirection, CRUD, file upload) immediately after implementation, followed by bulk-data testing using seeded and CSV-imported records to validate behaviour at scale.

4.9 Expected Outcomes

By the completion of this project, the following outcomes were achieved:

- A fully functional, role-based user management web application running on a local WAMP server.
- Secure registration and login flows with properly hashed passwords and no plain-text credential storage.
- A working admin panel supporting complete CRUD operations on user records, including profile picture upload.
- Practical, applied understanding of PHP sessions, prepared statements, MySQL schema design, and server-side authorization — skills directly transferable to internship and placement interview scenarios.

4.10 Certificates of Completion and Communication Proof

Offer letter:



FINZ FINTECH PRIVATE LIMITED

+91-9289310672

support@finz.live

https://finz.live/

INTERNSHIP OFFER LETTER

Date: 12-05-2026

To
APOORVA GUPTA,
IILM UNIVERSITY GREATER NOIDA

We are pleased to offer you an internship opportunity at FinZ Fintech Private Limited ("Company"/ "PW"), Corporate - Noida - UP (KLJ Noida One, First Floor, Sector 62, Noida, 201309) in FinZ. You will be joining the **Development** department in the role of **Full Stack Developer**. Your internship shall commence on **18-05-2026** ("Joining Date") and shall be for a period of **3 months** ("Internship Term")

Compensation

You'll be paid 0/- on a monthly basis ("Stipend").

You will not be under the direct payroll of the Company and therefore, you will not be receiving any of the employee benefits including, but not limited to, health insurance, etc. that the Company offers to its permanent and full-time employees.

For this internship, your major duties will include, but are not limited to activities and innovation.

It is understood that your internship is voluntary and treated as 'at-will'. Further, please note that your internship will be subject to the correctness of all the information and necessary documents furnished by you. In the event, it is found that any such material information furnished by you, whether verbally or in writing, at any time, is incorrect, misrepresented or fabricated, the Company shall have the right to terminate your internship without any notice or compensation.

During your tenure of internship, you would be governed by the Company policies and any other agreement that you may execute with the Company from time to time.

During your internship, you may have access to confidential, proprietary, and/or trade secret information belonging to the Company. All of this information shall be kept strictly confidential, and you shall not use it for your own purposes or shall not disclose it to anyone outside the Company. Upon conclusion of the internship, you shall immediately return to the Company all of its property, equipment, and documents provided to you during the term of your internship.

Plot No. B-8, Tower A, Noida One, Ground Floor, Sector 62, Noida, UP - 201309



FINZ FINTECH PRIVATE LIMITED

+91-9289310672
support@finz.live
<https://finz.live/>

During the term of your internship, you may create work reports, summaries, synopsis, projections among others for the Company, and all such work that is reduced to fixed form or is otherwise capable of any intellectual property protection will be the absolute property of the Company. You agree not to make claims over the same and to further do all acts, deeds or things as the Company may desire to enable the Company to have such rights over the works and or the work products.

As communicated to you at the time of offer, please ensure that effective immediately, you are required to work on your own personal laptop and no laptop shall be provided to you by the company for the purpose of fulfilling your work responsibilities and tasks. Furthermore, failure to comply with this requirement may result in leave without pay being applied for the duration of the non-compliance. Additionally, please note that if you are in non-compliance for a continuous period of 7 days, it may lead to the termination of your internship with the company.

In accordance with the terms of this internship offer letter, the internship period is anticipated to conclude after **3 months** starting from the **18-05-2026**. Nevertheless, should you desire to prematurely terminate your internship, you shall be obligated to provide the company with a written notice of 30 (thirty) days in advance.

To confirm your acceptance, please reply to the offer letter email with the following statement as a token of your acceptance: "I acknowledge all the terms & conditions and company policies, and I accept this offer."

We look forward to welcoming you to the FinZ team.

Yours Sincerely,
FinZ FinTech Pvt. Ltd.

Name: Piyush Jaiswal
Designation: National Head

Plot No. B-8, Tower A, Noida One, Ground Floor, Sector 62, Noida, UP - 201309



FINZ FINTECH PRIVATE LIMITED

+91-9289310672
support@finz.live
https://finz.live/

ACCEPTANCE OF THE INTERNSHIP OFFER LETTER

By signing this Internship Letter and thereby acknowledging and accepting this Internship Letter, you confirm and agree that such acceptance and signature constitutes a valid and legally binding Internship Agreement between you and the Company. You understand that by accepting and signing this Internship Letter, you are entering into a legally enforceable contract, and that you agree to be bound by all the terms and conditions set forth in this Internship Letter and all applicable Company Policies and Procedures.

For and on behalf of the "Intern"
Name: APOORVA GUPTA,
Date: 12-05-2026

Plot No. B-8, Tower A, Noida One, Ground Floor, Sector 62, Noida, UP - 201309

Certificate:



5. BIBLIOGRAPHY / REFERENCES

PHP Documentation — The PHP Group. Available at: <https://www.php.net/manual/en/>

MySQL 8.0 Reference Manual — Oracle Corporation. Available at: <https://dev.mysql.com/doc/refman/8.0/en/>

MDN Web Docs — HTML, CSS Reference. Mozilla Foundation. Available at: <https://developer.mozilla.org/>

W3Schools — PHP and MySQL Tutorials. Available at: <https://www.w3schools.com/php/>

phpMyAdmin Documentation. Available at: <https://docs.phpmyadmin.net/>

OWASP Foundation — Password Storage Cheat Sheet. Available at: <https://cheatsheetseries.owasp.org/>

Code Step by Step — YouTube Channel, PHP and MySQL Project Tutorials. Available at: <https://www.youtube.com/@CodeStepByStep>

Google Gemini — AI Assistant, used to generate sample CSV test data for database seeding. Available at: <https://gemini.google.com/>